

CYBERBULLYING DETECTION IN SOCIAL MEDIA CHATS

A REAL-TIME TOXICITY FILTER USING DEEP LEARNING

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Abstract

Online platforms, particularly massive multiplayer gaming communities and social media networks, frequently face significant challenges due to toxic behavior and bullying in chat environments. Cyberbullying not only detrimentally affects the mental health of users, leading to anxiety, depression, and social isolation, but also significantly reduces user engagement and retention on these platforms. Traditional moderation techniques, such as manual reporting or keyword-based filtering, have proven insufficient against the evolving landscape of internet slang and multimodal harassment. This project aims to build a comprehensive Real-Time Detection System that automatically flags, reports, or mutes abusive messages instantly. By combining advanced Natural Language Processing (NLP) models such as Bidirectional Encoder Representations from Transformers (BERT) for textual analysis with Deep Learning techniques for visual content (images, stickers, GIFs), the system provides a holistic approach to toxicity detection. This research paper details the methodology, system architecture, mathematical foundations, and experimental results of the proposed system, demonstrating a high classification accuracy of 96% and a precision of 0.94. The results suggest that deep learning-based approaches significantly outperform traditional machine learning baselines in complex gaming environments.

Keywords— Cyberbullying, BERT, NLP, Deep Learning, Toxicity Detection, Social Media, Convolutional Neural Networks (CNN), Attention Mechanisms.

1. INTRODUCTION

The exponential growth of digital communication has revolutionized how humans interact, bridging geographical gaps and fostering global communities. However, this connectivity has also introduced the pervasive issue of cyberbullying. Unlike traditional bullying, which is often limited by physical proximity and school hours, cyberbullying can occur 24/7, reach a

vast audience instantly, and provides the aggressor with a shield of anonymity.

Platforms such as Discord, Twitch, in-game chat systems (e.g., League of Legends, Valorant), and social media are often plagued by toxic behavior. This ranges from hate speech and threats of violence to persistent harassment based on identity, race, or gender. The impact of this behavior is severe. Victims often experience acute psychological distress, while platform administrators struggle to maintain community standards due to the sheer volume of data generated every second.

Manual moderation is no longer feasible at scale. A popular game might generate millions of chat messages per hour, making human oversight impossible. Furthermore, simple automated tools like "bad word lists" (keyword filtering) are ineffective. Bullies easily bypass these filters using misspellings (e.g., "k!!! yourself"), Leetspeak, or by using words that are benign in isolation but toxic in context.

This project, "Cyberbullying Detection in Social Media Chats," proposes a robust solution that leverages state-of-the-art Artificial Intelligence. Unlike simple keyword filters, our system employs the Bidirectional Encoder Representations from Transformers (BERT) model to understand the semantic meaning of text. Furthermore, recognizing that modern harassment often involves visual media, we integrate a Deep Learning module (CNN) to scan images and GIFs for offensive content. The primary objective is to create a real-time toxicity filter that enhances user safety without compromising the user experience or flagging innocent competitive banter.

2. LITERATURE REVIEW

The domain of automated text classification and toxicity detection has evolved significantly over the past decade, moving from rule-based systems to advanced neural architectures.

2.1. Early Approaches: Keywords and Rules

Early approaches primarily relied on Keyword-Based Filtering. These systems utilized static "blacklists" of offensive words. While computationally inexpensive, they suffer from high False Negative rates (failing to

catch abuse that uses creative spellings) and high False Positive rates. For example, a discussion about "killing a boss" in a video game would be flagged by a naive filter, frustrating users.

2.2. Traditional Machine Learning

Subsequent developments introduced Traditional Machine Learning models such as Naive Bayes (NB), Support Vector Machines (SVM), and Random Forests. These models utilize "Bag-of-Words" (BoW) or TF-IDF (Term Frequency-Inverse Document Frequency) features.

- **Naive Bayes:** Often used as a baseline due to its speed. However, it treats features independently, failing to capture the relationship between words.
- **SVM:** Proved effective for high-dimensional text data but struggles with scalability and capturing semantic nuance in informal internet slang.

While these offered improvements, they often struggled with the limitations of context. For instance, the phrase "This is a sick game" implies praise in gaming slang, but a model trained on general datasets might interpret "sick" as negative.

2.3. Deep Learning and Embeddings

The introduction of Word Embeddings (Word2Vec, GloVe) allowed models to capture semantic similarity. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks utilized these embeddings to process sequences of text, capturing context from previous words. However, LSTMs struggle with long-range dependencies and are sequential, making training slow.

2.4. The Transformer Era

The introduction of the Transformer architecture by Vaswani et al. (2017) and the subsequent release of BERT by Devlin et al. (2018) revolutionized NLP. BERT's ability to read text bidirectionally (simultaneously looking at words before and after the target word) using the "Self-Attention" mechanism allows for a deep understanding of linguistic context. This makes it superior for tasks involving sarcasm, slang, and implicit threats, which are common in cyberbullying.

3. THEORETICAL FRAMEWORK

3.1. BERT Architecture

The core of our text classification engine is BERT (Bidirectional Encoder Representations from Transformers). BERT is designed to pre-train deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers.

The architecture relies on the Attention Mechanism, specifically Scaled Dot-Product Attention, which can be described mathematically as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where:

- Q is the Query matrix.
- K is the Key matrix.
- V is the Value matrix.
- d_k is the dimension of the keys.

This mechanism allows the model to assign different "weights" or importance to different words in a sentence relative to the word being processed. For toxicity detection, this means the model can understand that "I will kill you" (high attention on the verb-object relationship) is a threat, whereas "I killed the process" (attention on technical context) is not.

3.2. Convolutional Neural Networks (CNN)

For visual analysis, we utilize Convolutional Neural Networks. CNNs are specialized for processing grid-like data such as images. The core operation is the convolution, where a filter (kernel) slides over the input image to create feature maps:

$$(I * K)(i, j) = \sum_m \sum_n I(m, n) K(i - m, j - n) \quad (2)$$

These feature maps detect edges, shapes, and textures. In deeper layers, they detect complex objects. For our purpose, the CNN is trained to recognize visual patterns associated with NSFW content or hate symbols used in memes.

4. PROPOSED METHODOLOGY

To overcome the limitations of existing systems, we propose a hybrid, multi-modal architecture. The system processes two distinct data streams: textual chat logs and visual media attachments.

4.1. System Architecture

The application is structured around five modular components to ensure scalability:

1. **Data Preprocessing Module:** Raw data from chat inputs is noisy. This module performs essential cleaning steps:
 - **Normalization:** Converting text to lower-case.
 - **Noise Removal:** Removing HTML tags or system artifacts, but specifically *retaining* punctuation often used in emoticons (e.g., ":", ":(").

- **Tokenization:** Breaking sentences into WordPiece tokens compatible with BERT.

2. **Text Classification Module:** This module receives the clean tokens and feeds them into the BERT model. It utilizes a softmax classifier on top of the BERT output to predict probability scores for six labels: Toxic, Severe Toxic, Obscene, Threat, Insult, and Identity Hate.
3. **Visual Detection Module:** Running in parallel, this component analyzes media. It resizes images to 224×224 pixels and passes them through a pre-trained VGG-16 or ResNet model fine-tuned for content safety.
4. **Chat Simulation Interface:** A frontend application mimicking a gaming chat UI was developed for demonstration. It connects via WebSocket to the backend for low-latency classification.
5. **Reporting & Logging Module:** Flagged messages are logged in a secure Firestore database. This allows administrators to review false positives and update the model.

5. IMPLEMENTATION DETAILS

5.1. Hardware and Software Setup

The model training and inference were conducted using the following configuration:

- **Language:** Python 3.8
- **Frameworks:** TensorFlow 2.0, Keras, Hugging Face Transformers.
- **Compute:** NVIDIA Tesla T4 GPU (via Google Colab Pro) for training acceleration.
- **Frontend:** React.js for the chat simulation.

5.2. Datasets Used

Two primary datasets were employed to ensure the model's robustness:

1. **Jigsaw Toxic Comment Classification Dataset:** This dataset contains approximately 159,000 Wikipedia talk page comments, labeled by human raters for toxic behavior. It serves as the foundational training set.
2. **Cyberbullying Detection Dataset:** A smaller, domain-specific dataset containing tweets and gaming chat logs, used to fine-tune the model for social media slang.

5.3. Training Parameters

The BERT model (specifically bert-base-uncased) was fine-tuned with the following hyperparameters:

- **Batch Size:** 32
- **Learning Rate:** 2×10^{-5} (Adam Optimizer)
- **Epochs:** 4 (Early stopping applied to prevent overfitting)
- **Max Sequence Length:** 128 tokens

6. EXPERIMENTAL RESULTS

The performance of the system was evaluated using a hold-out validation set comprising 20% of the original data. We evaluated the model based on Accuracy, Precision, Recall, and F1-Score.

6.1. Performance Metrics

- **Accuracy:** The percentage of total predictions (both toxic and non-toxic) that were correct.
- **Precision:** The ability of the classifier not to label a negative sample as positive (Low False Positives).
- **Recall:** The ability of the classifier to find all the positive samples (Low False Negatives).

Table 1: Performance Metrics of the Proposed System

Metric	Value	Interpretation
Accuracy	96%	High overall correctness.
Precision	0.94	Minimal false flagging.
Recall	0.83	Captures most abuse.
F1-Score	0.88	Balanced performance.

6.2. Comparative Analysis

We compared our BERT-based approach against baseline models trained on the same dataset to validate the efficacy of using Transformers.

Table 2: Comparison with Baseline Models

Model	Accuracy	F1-Score
Naive Bayes	78.4%	0.62
SVM (TF-IDF)	84.1%	0.71
LSTM	91.2%	0.81
Proposed (BERT)	96.0%	0.88

As shown in Table 2, the BERT model significantly outperforms simpler statistical models and offers a substantial improvement over LSTM networks. The Naive Bayes model particularly struggled with sarcasm and negated sentences (e.g., "I am not happy" vs "I am happy"), which BERT handled correctly due to its attention mechanism.

6.3. Visual Detection Performance

The CNN module for image detection achieved an accuracy of 89% on a test set of 1,000 images containing a mix of safe gaming screenshots and offensive memes. While effective, it struggled slightly with text embedded inside images (OCR tasks), which is a noted area for future improvement.

7. DISCUSSION

7.1. Handling Context and Slang

One of the most significant challenges in gaming chat analysis is distinguishing between toxic behavior and "trash talk" or competitive banter. Phrases like "I killed you" or "shoot him" are standard in First-Person Shooter (FPS) games. A rigid rule-based system would flag these immediately.

Our results indicate that the BERT model, having been pre-trained on a massive corpus of English text, possesses a degree of contextual awareness. When fine-tuned on gaming logs, it learned to lower the toxicity probability for game-specific violence terms unless they were accompanied by personal insults or racial slurs.

7.2. Ethical Considerations and Bias

Deploying AI for moderation introduces ethical concerns.

- **Bias:** AI models can inadvertently learn biases present in training data. For instance, discussions about certain minority groups might be flagged as toxic simply because those keywords frequently appear in toxic threads in the training data. We mitigated this by using the "Identity Hate" label explicitly and balancing the dataset.
- **Privacy:** The system requires analyzing user messages. To protect privacy, our architecture supports "stateless" processing, where messages are processed in memory and discarded immediately unless flagged as a policy violation.

8. CONCLUSION

This project successfully designed and implemented a "Real-Time Toxicity Filter" capable of detecting cyberbullying in social media chats. By moving beyond archaic keyword blacklists and embracing modern AI architectures like BERT and CNNs, the system achieves high accuracy and contextual awareness.

The solution not only identifies textual abuse but also screens visual content, providing comprehensive protection. With an accuracy of 96%, the system demonstrates that automated tools can be both effective and reliable, potentially serving as a critical tool for platform administrators in the fight against online

harassment. The high precision score (0.94) is particularly vital for user acceptance, as it minimizes the frustration caused by false accusations.

9. FUTURE SCOPE

While the current system is robust, there are several avenues for future enhancement:

1. **Multilingual Support:** Currently, the model is optimized for English. Future work involves training on multilingual datasets (like mBERT or XLM-RoBERTa) to support global platforms and code-switching (mixing languages in one sentence).
2. **Voice Chat Analysis:** Gaming is increasingly moving toward voice communication. Integrating Speech-to-Text (STT) modules to analyze voice chats in real-time would be a valuable addition, though it presents challenges regarding latency and privacy.
3. **Video Analysis:** Expanding the visual module to analyze video streams (live streaming) for inappropriate content frame-by-frame.
4. **Edge Deployment:** Optimizing the model using techniques like quantization or knowledge distillation to run directly on user devices (mobile/browser). This would reduce server costs and improve privacy by keeping data on the client side.

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